BIOCHEMISTRY CUMULATIVE EXAM

February 25, 2000

The cytochrome bc_1 complex is an integral mitochondrial protein which transfers two electrons from ubiquinol, QH_2, with a redox potential of 0.04 V, to two molecules of cytochrome c, with a redox potential of + 0.26 V. This electron transfer process is coupled to the translocation of 4 protons from the inside to the outside of the mitochondrial membrane. The cytochrome bc_1 complex contains four redox centers, cytochrome b_L, cytochrome b_H, the Rieske iron-sulfur protein, and cytochrome c_1. Under active electron transfer conditions, the electrical potential across the mitochondrial membrane was measured to be $V_{out} - V_{in} = +0.1$ V, and the pH inside the mitochondrial membrane was $pH_{in} = 8$, and $pH_{out} = 6$.

1. (25) Give a simple diagram showing how the cytochrome bc_1 complex couples the energy of electron transfer to pumping protons across the mitochondrial membrane. What is this process called?

2. (25) Calculate the free energy ΔG^o for electron transfer from 1 molecule of QH_2 to 2 molecules of cytochrome c.

3. (25) Give a formula for the free energy change ΔG for pumping protons from the inside to the outside of the mitochondrial membrane, and calculate ΔG for proton pumping by the cytochrome bc_1 complex under the above conditions.

4. (25) Calculate the efficiency with which the cytochrome bc_1 complex couples the exergonic free energy of electron transfer to the endergonic free energy of proton pumping.

Possibly useful formulas:

$\Delta G^o = - RT\ln K_{eq}$; $\Delta G^o = nF\Delta E_o$;
$\Delta G = \Delta G^o + RT\ln\{\text{products/reactants}\} + nF\Delta V$
F = Faraday's constant = 23 kcal volt^{-1}; R = 1.987 cal mol^{-1} deg^{-1}